

CONTENTS

Foreword ix

Introduction: From Scarcity to Prosperity within a
Generation 1

PART ONE **SCARCITY**

- 1 The Failure of Money: The Competitive Society 11
- 2 The Myth of Money: What It Really Is 23
- 3 A Fate Worse Than Debt: Interest's Hidden Consequences 37

PART TWO **PROSPERITY**

- 4 The Flying Fish: A New Perspective on Money 57
- 5 The Future Has Arrived But Isn't Distributed Evenly . . .
Yet! 73
- 6 Strategies for Banking 95
- 7 Strategies for Business and Entrepreneurs 119
- 8 Strategies for Governments 141
- 9 Strategies for NGOs 159

PART THREE **RETHINKING MONEY**

10	Truth and Consequences: Lessons Learned	175
11	Governance and We, the Citizens: An Ancient Future?	187
12	Our Available Future: The Cooperative Society	203
13	Rethinking Money: From Scarcity to Sustainable Abundance	215
	<i>Notes</i>	225
	<i>Bibliography</i>	241
	<i>Acknowledgements</i>	249
	<i>Index</i>	251
	<i>About the Authors</i>	261



Chapter Eight

STRATEGIES FOR GOVERNMENTS

Money is the crowbar of power.

FRIEDRICH NIETZSCHE,
19th-century German philosopher

Trash was a nightmare problem in Curitiba, the capital of the south-eastern state of Paraná, Brazil. There was a large slum population dwelling in shantytowns, makeshift, improvised housing constructed from corrugated steel, cinder blocks, and whatever else was available.

“Back in 1989, the primary problem we faced was garbage in the favelas. We needed to avoid pollution in our streams and, of course, to protect the kids who were playing in what were very contaminated areas. The problem was that we had to have the garbage collected with trucks, but they couldn’t get into the favelas because the pathways were too narrow and the terrain was very hilly,” recalls Jaime Lerner, who was mayor at the time.

The issue was further compounded because the city simply did not have the funds to deal with the crisis. Raising the necessary money through conventional methods, such as requesting funding from the federal government, was not an option. Something else had to be done.

This dilemma facing governments at whatever level of authority is commonplace some three decades later. Even in better economic times, making the limited conventional currency stretch has always

been troublesome. With tax revenues to fund programs and services dwindling today because of the harsher economic climate, everyone is feeling the pinch. National governments have had to slash programs, particularly in Europe, following a series of austerity measures to get spending under control.

Lerner, however, took stock of the resources on hand. There was an abundance of food, given the region's rich farmland, proximity to the sea, and subtropical climate. Additionally, there was an underutilized municipal bus system and people with a lot of time on their hands. Lerner, who was trained as an architect and consequently thought in terms of systems and their integration, had an idea!

It was a strategy that developed over time. Large metallic bins were placed on the streets at the edge of the favela neighborhoods. Lerner recalls, "We separated the trash into three unique components: one third, such as paper, glass, and cartons, was commercially recyclable; one third was biowaste (available to produce organic compost); and the other third was trash for the landfill. Those who collected and sorted the trash were given tokens to ride the local bus system."

Thousands of children responded by picking the neighborhoods clean. The children have taught their parents how to do it. Lerner noticed that to earn bus tokens, some slum dwellers even collected and sorted garbage along the highways, making it easier to get the waste picked up by trucks.

People made use of the tokens to travel downtown, often to find work. The bus tokens were soon accepted at local markets in exchange for food. The project expanded from there. In one three-year period, more than 100 schools traded 200 tons of garbage for 1.9 million notebooks. The paper-recycling component alone saved the equivalent of 1,200 trees—each day!¹

Eventually, more than 70 percent of Curitiba households became involved in the various programs. The 62 poorer neighborhoods alone exchanged 11,000 tons of garbage for nearly a million bus tokens and 1,200 tons of food. Other programs were created to finance the restoration of historical buildings, create green areas, and provide housing,

all by methods that didn't create any financial burden on the municipality.

Thanks to Lerner and his innovative methods, Curitiba has made major strides in other social sectors, providing the city with some of the best quality-of-life indices in the world. Sixty to 70 percent of Curitiba's trash is recycled in situ, possibly the highest percentage in the world.

The experience gained, along with the lessons learned as mayor of the city of Curitiba, served Lerner well when he later was elected governor of the state of Paraná. The ideas and their implementation served as a blueprint on a larger scale.

Lerner adds, "When I was a governor, from 1994 to 1998 and again from 1998 to 2002, originally we wanted to avoid loans, as in the case of Argentina and the \$800 million World Bank loan they took out to clean their bays. We decided that it's not only a question of money but also a question of mentality. It's not only a question of public works but also it's how to successfully stimulate a learning process. So we didn't have the money to clean our bays. So instead we made agreements with our fisherman. When they catch the fish, the fish belong to them. When the days weren't good for fishing, they catch garbage, we pay for the garbage with our tokens. The more garbage they fished, the cleaner the bay became; the cleaner the bay became, the more fish they could catch. It's a win-win solution."

Another major problem is street children. "These children were usually begging for money to feed their families. We tackled that problem by making an agreement with each family: We provided each of them with a monthly basket of food for as long as they kept their children in school."

The idea was to implement this program across all of the 399 municipalities of Paraná, which is home to over 10 million people.

"When we realized the low cost of monthly baskets of food, and how they helped those living in poverty by freeing up their limited resources to pay for other things, everybody got very excited. Instead of having to use all their money to buy food, they could now use the



Garbage that is money. When the fish aren't biting, the fishermen clean the bay instead.
Photo credit: Instituto Jaime Lerner.

money in different ways and at the same time be sure that their kids were going to school,” Lerner recalls.

The many initiatives—environmental cleanup, city restoration, job creation, improved education, disease intervention, and hunger prevention—were each tackled with various cooperative currency systems. This all happened without having to raise taxes, redistribute wealth, issue bonds, rely on charity, or obtain loans from the federal government or organizations such as the World Bank and the International Monetary Fund (IMF). The improvements burdened no one. Everyone benefited.

Curitiba, along with the state of Paraná, discovered a means by which to match unmet needs with unused resources. In 1990, Curitiba was honored with the United Nations Environmental Program's (UNEP) highest environmental award.² Curitiba was awarded the Globe Sustainable City Award in 2010.³ This award recognizes cities and municipalities around the world that excel in sustainable urban development.

None of the exceptional features of what has happened in Curitiba would have been possible without the various cooperative currency systems implemented by Jaime Lerner. Curitiba is the first city to have systematically used a “monetary ecosystem.” It is therefore a practical demonstration of what becomes possible within less than one generation when one rethinks money. There have been several PhD studies dedicated to the innovations in city planning and public transport systems of Curitiba. What is surprising is that, notwithstanding the availability of 25 years of data, no systemic economic study has been performed on Curitiba.

THE CIVIC: ECONOMIC STIMULUS WITHOUT DEBT

In the case of Curitiba, the results of using a cooperative currency in purely economic terms are worth noting. From 1975 to 1995, the GDP of the city increased by 75 percent more than the rest of the state of Paraná and by 48 percent more than Brazil’s as a whole. The average Curitibaano earned more than three times the country’s minimum wage. If nontraditional monetary gains, such as the exchange of garbage for provisions, are taken into consideration, the real total income for residents was at least 30 percent higher still. The results in human terms—in the renewal of dignity and hope for a better future—are incalculable.⁴

This was accomplished by rethinking money, moving from a model of scarcity that the conventional system inadvertently encourages to one of prosperity with the use of smartly designed cooperative currencies.

Traditional economic stimulus is an effort by government to boost an ailing economy by pumping more money into it, usually by cutting taxes and by borrowing money. With these measures, it is hoped that liquidity in the system will increase, which in turn will lead to greater financial resilience, as there is now money for job creation and/or other programs. But this approach has serious negative repercussions: Both increase the national debt. Lately, the net effect has been a tightening

of the proverbial belt with drastic austerity measures, such as slashing social programs. This has led to massive civil unrest, particularly in Spain and Greece.

Richard A. Epstein, a senior fellow at the Hoover Institute writes, “Grim is the right word to describe the latest economic news from both the European Union and the United States. Throughout the European Union, austerity programs have failed both politically and economically. In Spain, unemployment rates have soared above 24 percent. The Dutch government is on the edge of collapse because of the popular and political unwillingness to accept the austerity program proposed by its conservative government. Romania is not far behind. Greece, Italy and Portugal remain in perilous condition. . . . On the American front, the decline of GDP growth to 2.2 percent rightly raises fears that our sputtering domestic recovery is just about over.”⁵

There is a solution, as the events in Curitiba illustrate. Government, whether at a local, state, or regional level, can take its economic fate into its own hands and issue a cooperative currency. This cooperative medium of exchange, which can be customized to any given government’s needs, is a currency called the *civic*. The key component of such a solution is that it provides a financial Keynesian stimulus with a fundamental and critical difference: It doesn’t generate any additional debt.

Indeed, a Keynesian stimulus means that government steps in and launches projects that it funds through deficit spending. The government then borrows from the banking sector, going further and further into debt. This is because the government cannot issue itself the currency. In contrast, if the government launches the same types of projects but issues cooperative currencies to fund them and requires taxpayers to make a contribution payable only in that same currency, no debt is generated.

Furthermore, the process is more bottom-up than the usual top-down central government stimulus plans. It tends to be more successful since the cash injection can be very precisely applied. Residents and local government members clearly have a better insight into a

community issue with all its nuances than does a central government bureaucracy. And since the currency's governance needs to be more transparent and easier to manage because of its size, there is less opportunity for fraud and misappropriation. In Chapter 11 the importance of governance issues are addressed.

HOW IT WORKS

A yearly contribution in civics would be requested of all town or city households by their local government authority. Residents would earn civics, which would be issued by the city, by participating in activities that contribute to the city's democratically agreed-upon goals and objectives. Agreement as to the nature of these tasks might be reached by canvassing neighborhoods door-to-door or by online voting. This unit of account could be one hour of time, valued at the same rate for everybody. So, for instance, if a city aspires to be greener and more eco-friendly, the activities could include growing food on terraces, rooftops, or vacant plots of land; taking responsibility for plants and trees in the neighborhood parks; or training people in city-based horticulture.

From a purely economic angle, if an annual tax of \$1,000 can be replaced with 10 hours of civic activity per household, anyone earning less than \$100 per hour should be interested in joining the system. Additionally, the civics system has the added benefit of allowing people to earn an income from civil activities.

Such an approach provides, in fact, a decentralized Keynesian stimulus at the city scale without creating new debt or incurring further costs to the government. Contrast this to what's happening now, as governments try to keep up the social system with the euro, which clearly isn't working.

There is no obligation to personally perform any of the tasks rewarded in civics. To begin, families with children, infirm, or individuals with special needs would be exempt. There would be two other ways of opting out of participation. The first could be by paying an extra amount in dollars as part of one's annual taxes (e.g., \$1,000 for

the example above). The second option for people not interested or without the time to personally perform any of the tasks would be to purchase civics via an online market, openly and transparently. People who earned more civics than they needed for their annual contribution could sell them in that market. The buyer of the civics could make the purchase in conventional money or as an exchange for any good or service acceptable to both parties.

The government's role should be to ensure that fake civics are not in circulation and that exchanges are transparent and fair. The government should not tie the value of the civic to the national currency. If the government wants the value of civics to rise relative to the national currency, the most effective way would be to require a bigger contribution payable only in civics and, accordingly, if it wanted the value to drop, it could require a smaller contribution to be paid in civics.

Furthermore, the process can be targeted to specific population segments and should be countercyclically tuned to local conditions. For instance, specific programs that pay in civics could be implemented for young people when their unemployment level is abnormally high, as is the case now in parts of Michigan and California, for example, or in Portugal, Ireland, Greece, and Spain in Europe.

The legitimacy of the civic completely depends on the legitimacy of the process by which the choices are made on the activities that can be awarded civics. Bottom-up democratic processes are critical to the genuine success of what is proposed here. This leads to the critical question of appropriate governance for such systems, as will be shown in Chapter 11.

NATIONAL CURRENCY CRISIS SOLUTION?

At the core of our current financial predicament is the issue that the banks aren't lending, and it's not because businesses in general aren't viable. Cooperative currencies, however, can remedy this cash crunch.

In terms of a national crisis, a C3 currency, as explored in Chapter 7, could be used to take care of some of the commercial business currently conducted in national currencies. In a case for the eurozone, for example, countries like Spain, Portugal, Greece, or whichever is in trouble could continue using the euro currency for everything having to do with international activities: tourism, shipping, importing, and exporting. Their respective governments could, in addition, create a new version of the peseta, escudo, or drachma to be used for internal social and environmental businesses. This neonational currency essentially could be spent into existence by the government itself, for specific purposes, without incurring debt in the financial system.

Thomas Mayer, former chief economist for Deutsche Bank, made in July 2012 a proposal for Greece that is surprisingly radical for his background. It acknowledges that the great majority of Greeks don't want to leave the euro and that they don't agree with the extreme austerity program imposed in Greece.

Mayer's solution is called the *geuro*, a second currency that would circulate in Greece in parallel with the euro and, Mayer says, "solve all of Greece's problems." For the first time, a mainstream economist, working for a major bank, is endorsing the idea of a complementary currency!

The geuro would be used immediately in most domestic transactions. For the purchase of essential imports, geuros would have to be exchanged for euros, "most likely at a hefty discount of 50 percent or more. Since an increasing number of domestic goods, services, and wages would be paid in devalued geuros, the export sector could reduce its prices in euros and regain competitiveness against foreign suppliers."⁶ The main difference between the geuro and the civic is that the geuro would be issued as a central government IOU and the civic would be issued by cities or regions without generating additional debt. But the civic would otherwise provide a mechanism to provide the same benefits as the geuro in terms of gradually rebuilding the competitiveness of Greek labor.

Edgar Kampers, director of the Dutch Qoin, has been working in the domains of sustainable economic development and cooperative community currencies since 1998. He remarks, “Current legal tender currency systems are very strong in developing high profits and building a globalized society. They have proved less effective, however, in supporting regional economic development, stimulating ecological policy goals and behaviors, and encouraging an active civil society. The current economic and financial crisis faced by many regions in Europe calls for new arrangements for communities to remain or become resilient.”⁷

In the province of Limburg in Belgium, for example, there was a waste cooperative between all the municipalities in the province, and this organization calculated the difference in price between new kinds of campaigns. “What they found was if they do a traditional awareness campaign, then they have a response of less than 2 percent. If they do the same thing with the same costs per person in that municipality, and they use a cooperative currency system, then they have a response of almost 40 percent. That’s 20 times better! It’s the only successful way of changing the behavior of the people on a limited municipal budget. And coupled with the information society, which makes computing, smart phones, tablets, and this type of technology so cheap and easy, it is now available everywhere,” Kampers added.

Lisa Conlan, CEO of TimeBanking USA, notes, “Given the current credit crunch that states and municipalities are experiencing, there is no reason why a wide variety of services could not be paid for in a local currency, such as time dollars. There’s snow removal, garbage collection, even community gardening.”⁸

A rich tapestry of initiatives comes to light with the understanding that unused resources can be linked with a whole host of unmet needs. Such exchanges provide solutions for cash-strapped charities and government bodies at different social levels, resulting in huge economic and social benefits.

TOREKES

Rabot is an immigrant district in Ghent, the fourth-largest city in Belgium. Rabot is the poorest community in the entire region. Most of the population lives in low-income apartment buildings in one of the most densely populated localities in Europe. Well over 20 languages are spoken, with Turkish the most prominent. Rabot suffers from high unemployment and the usual symptoms of urban decay, which have profound effects, both physical and metaphysical.

In 2009, one of the authors was asked what could be done to improve the area. The starting point was a survey to find out what residents wanted for themselves. Many, particularly those living in high-rise apartment buildings, wanted to have access to a few square yards of land for gardening, growing vegetables, and flowers.



Ghent, Belgium. Residents pay for their plots of land exclusively in the town's local currency—torekes. *Photo credit:* Stefanie Overbeck.

The city owned land in the neighborhood, including a site where an old factory had been demolished and the land was left abandoned. Rather than leaving it derelict, the land was divided into plots measuring four square meters each, and these plots became available to rent on an annual basis, at the cost of 150 *torekes*, a newly introduced local cooperative currency. Torekes are available only in paper form for this pilot project, on the request of the participants themselves. One of the reasons is simplicity.

To earn *torekes* (Flemish for “little towers,” emblematic of the neighborhood), a long list of urban agricultural improvements and beautification activities could be done. Participants were rewarded for activities such as putting flower boxes on the windowsills facing the street, planting and maintaining sidewalk flower containers, placing “no advertising” labels on mailboxes to reduce junk mail, or helping to clean up a neighborhood sports field after a game. In all, 526 different activities were performed.

Wouter Van Thillo, who has been managing the project since 2011, describes the series of events that ensued: “In addition to paying for the rent of the neighborhood gardens, arrangements were also made with the local shops to accept *torekes* for specific goods that the city wants to encourage, such as low-energy lamps and other green products, or fresh seasonal vegetables. Participating shops can either use the *torekes* for their own participation in local activities or simply get them reimbursed for euros at the city office. *Torekes* can also be exchanged for public transport tickets and seats for cultural events or movies. All activities with very low marginal costs for additional participation, such as buses and cinema, are ideal for this scheme.

“We’ve also found, as this program has gained traction, that well-to-do residents from other parts of Ghent are now participating because it’s fun.”

This simple paper currency with no demurrage is considered to be a major success. Not a week goes by without visitors from other towns and cities across Europe stopping by to learn from this example. What this experiment has proven is that it produces a much broader social

impact with the same euro budget, providing effective leverage in using conventional currency. Specifically, even at this very early pilot scale, conservatively three times more activities have been produced with the same amount of euros. If the implementation was fully scaled up, it has been estimated that this multiplier could potentially rise to a factor of 20.

These two colorful models of municipal and state governments finding inventive ways to resolve issues in their respective communities stimulate questions on how these methods can be expanded and improved. For the foreseeable future, government at all levels will continue to grapple with serious budget shortfalls, thus obliging cuts not only in discretionary programs but also to critical services and benefits. A city-issued currency such as *torekes* makes it possible to multiply the results achieved with a given budget in conventional money.

EDUCATION

The cost of education is a burden usually carried at the federal level of government. The seed of a provocative response to budget concerns was planted over a decade ago in Brazil, when the education fund had grown to more than \$1 billion. Conventional solutions—such as the very successful GI Bill approach of the United States that funded education for veterans after World War II—would provide student loans directly to individuals for their own education. Another idea was crafted by one of the authors in cooperation with Gilson Schwartz, an economics professor at the University of São Paulo, Brazil.

The framework of what has since matured into the Creative Currencies Project was to leverage educational funding by using a multiplier that would allow many more students to benefit from the same amount of money with some surprising results.

The first model was the *saber* (meaning “to know” in Spanish and Portuguese), a specialized education paper currency allocated to primary and secondary schools, particularly in economically depressed areas. This currency is first given to the youngest students (7-year-olds),

who transfer it to older students (for instance, 10-year-olds) in return for tutoring help with their schoolwork. The 10-year-olds can then do the same thing with 12-year-olds in compensation for the hours spent mentoring, and the latter with 15-year-olds and so on. At the end of this learning chain, the sabers would go to 17-year-olds, who could use them to pay for part or all of their university tuition.

The saber offers a number of advantages. It circulates among students at different grade levels, the last of whom go to college; it promotes better education through learning by teaching; it offers a better chance for students to go to a university; and it allows educational funds to be used for many more students than if they were used for just one scholarship, thereby creating a learning multiplier. For instance, the university would be able to exchange sabers for conventional money through the education fund but at a discount of, say, 50 percent. This makes sense because most of the costs at a university are fixed and, similar to empty seats on an airplane, the marginal expense of an additional student has little impact on those costs.

The total learning multiplier per dollar for the education budget allocated to this project would, therefore, be a factor of 10 (five times for the exchanges among students of different ages, multiplied by two for the arrangement between the education fund and the university, which discounts the saber by 50 percent). This 10-fold learning multiplier was calculated in terms of the financial effect of \$1 billion spent through the saber system, delivering a total of \$10 billion worth of education.

Another factor, learning retention rates, further increases this figure. Several decades of research have shown that learning retention depends less on the person or the topics involved than on the delivery system. What is striking is that our traditional educational system commonly uses the two least effective methods available: lecturing and reading, through which, respectively, only 5 and 10 percent of what is taught is retained. At the other end of the spectrum, an impressive 90 percent retention rate applies to whatever one teaches others!

So when the learning retention rates increase from 5 to 10 percent (normal educational methods) to 90 percent (teaching others) is factored in, another 10-fold multiplier effect develops. In other words, spending \$1 billion through the saber system could roughly be estimated to generate as much as \$100 billion worth of retained learning, in comparison to the conventional scholarship grants approach.

The saber program could also be expanded beyond the conventional classroom: eight-year-olds could teach their newly learned reading and writing skills to grandparents who are illiterate; students could help the elderly or handicapped by reading to them or recording their oral histories. These programs would encourage intergenerational relationships and further learning, not to mention creating extra assistance for the elderly without burdening governmental budgets.

Demurrage can be used to control the balance each year between the number of students wanting university seats and the availability of seats. A demurrage fee would be applied to encourage students to use sabers for the year that is printed on the saber itself, for example, 2014. If the sabers were not used to pay for tuition before or during that year, they could be exchanged for sabers of the following year—the year 2015—but with a 25 percent penalty. This creates a strong incentive to use sabers on or before their deadline.

“While the project hasn’t been adopted on a federal level in Brazil due to the vicissitudes of Brazilian politics, we have created instead two other currencies, *talents* for those involved in the practical application of knowledge, and *alegres* for activities that bring joy or make the world a more beautiful place,” adds Schwartz. Both use the same principles as the saber, in which young people teach other young pupils.

He continues, “We are now working with the Department of Finance and the police. Goods that have been seized because of unpaid taxes or contraband will be available for purchase by our students using any of the three creative currencies.”

WISPOS

Back again in the city of Ghent, Belgium, the local educational authority has been challenged by a variety of issues associated with the education of a large immigrant population, many of whom are illegal. Teachers report students having serious deficits in the rudiments of several subjects, the worst being French and mathematics. Falling behind academically undoubtedly contributes to being further marginalized in the future. Something creative to break the cycle had to be found.

Jeffrey Freed, author of *Right-Brained Children in a Left-Brained World*, a best seller and a classic in the field of education, has identified the general learning styles of the Millennials, children born since the turn of this century. “These kids have grown up in a world very different than previous generations. Just think about the impact of technology alone. Many tend to be more right-brained dominant in that they think in pictures. They process information in a nonlinear way. So in doing a math problem, for instance, they can get the right answer, but they can’t retrace their steps. They get bored and lethargic easily. The *drill and kill* method of teaching just won’t work. What succeeds is if they are taught in short time spurts. They need to understand the context of what is being taught, so learning in field trips and through other rich experiences is the way to go.”⁹

With this insight, one of the authors and Igor Byttebier, a resident of Ghent and a consultant to the innovative program, worked with the teachers of the Wispelberg School, one of Ghent’s high schools, to design an hour-long program with a class of 12-year-olds that takes place every two weeks. Kids from the senior class participate, too, by assisting the teachers. They designed games around learning so it would be fun and provide novelty. To make it even more interesting, they divided the class into teams that would compete against each other for the entire academic year. *Wispos*, a token currency, was awarded for correct answers and debited for incorrect ones.

Igor Byttebier explains, “French is a difficult language as the pronunciation is so subtle to the ear. So we devised a game called twister.

There's a large checkerboard on the floor in bright colors, five squares by five squares, 25 squares in all. The differently conjugated verb endings are placed in different squares. The teacher shouts out a phrase, and the team rep has to jump onto the right square. If correct, two wispos are given; if wrong, one is forfeited."¹⁰

On average, between 6 and 22 wispo tokens were won by each of the students, who could then purchase fruit and other healthy food items from the school store.

The project has been running for two years now, and, given its success, a second high school, KTA, has adopted the program. "When tested, the students scored 20 percent better than before as a result of the games, and in mathematics, the improvement in test scores was, on average, 10 percent."¹¹ Just as important is the interaction between students of first and last years, combined with physical and intellectual activity. This is an enormous help to their social integration and self-esteem," adds Byttebier.

Although this project is in its very early stages, it does serve to illustrate that linking an unused resource with an unmet need and introducing a new currency can bring about profound changes.

There are several key takeaways from all these examples. Besides cooperative money's capacity to engender greater collaboration and mutual support while providing cash-strapped governmental agencies with a way to leverage their tight resources, there is an even more powerful motivator at work. Traditionally, efforts to bring about a change in behavior have focused on increasing taxation (for example, increasing taxes on tobacco products to discourage smoking) and regulation, which tries to outlaw certain behaviors (such as Prohibition tried to do to alcohol consumption during the 1920s in the United States). The incentive stimulated by cooperative currencies, however, uses a carrot rather than the proverbial stick.

And perhaps more important, it demonstrates that it is possible to create economic stimulus without the negative effect of increasing debt.

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Chapter Nine

STRATEGIES FOR NGOs

*Money does not have to be legal tender.
It can be what one might call “common tender,”
i.e. commonly accepted in payment of
debt without coercion through legal means.*

RICHARD TIMBERLAKE, former Professor Emeritus of
Economics at the University of Georgia¹

Within the space of less than 10 years, the small rural village of Blaengarw, South Wales, with high unemployment and a bleak future, totally transformed itself. David Pugh, a local TimeBanking manager who oversees the currency explains, “On the Welsh index of material deprivation, we were 128th on the scale of 1,800 communities with one being the worst in the country. We’ve now progressed to 735th in just a decade. So we’ve climbed more than 600 places. These statistics are based on such things as antisocial behavior, crime, and of course, unemployment—in fact, there is a whole array of indices. We’ve really come a very long way.”²

The residents of Blaengarw worked their way out of the all-too-common story of social blight and decay by making an assessment of their unused resources and their unmet needs and, in this case, linking these with a time-banking currency coordinated by a regional and local NGO.

This former mining village of some 1,895 souls was trapped in a postindustrial depression, exacerbated perhaps by its remoteness in the Garw River valley. Although just physically 30 miles from Cardiff, the capital, it might as well be 3,000 miles away, as there is only one road in and out of town and a sporadic public bus service. David Pugh recalls, “The last pit closed in our village in December 1985. We were employing 7,000 men in the four pits in its heyday. Eighty percent of the local population was involved in mining in some fashion. So it was a disaster when all the mines closed down. Deprivation started setting in, as there was no formal employment replacing the mine. People made do with being on the dole [public assistance] or whatever casual work they could find. There wasn’t much, though. Furthermore, many people couldn’t work because they were disabled through coal dust inhalation and mining accidents. Many were physically incapable of doing any other work. We had to rebuild our village from there—from little or nothing.”³

Becky Booth is program coordinator and cofounder of Spice, which is part of the Wales Institute for Community Currencies. Spice fosters 40 time currencies in Wales and 12 in England. Booth says, “My research has been around postdisaster recovery and looking at the way the agencies that came into disaster situations work through the local community. My prime interest was around empowerment, participation, social management, and really looking at the role of NGOs in that context. I discovered that TimeBanking, along with other designs, is a very simple tool to enable the people to be genuinely involved in a totally different way with NGOs and public services.”⁴

Activities are centered around the town’s 100-year-old Miners’ Welfare Hall. The local activities are run by a community nonprofit organization, Creation Development Trust, which employs two workers to oversee the TimeBanking. For each hour of service given to the community, a credit of one hour is exchanged. In the first year, 150 people took part; there are now over 1,000 members and 30 groups; 15 new social enterprises have been established, and new learning opportuni-



The sculpture is a project completed by Blaengarw time credit members. *Photo credit:* Spice network.

ties have been created. Participants contribute 60,000 hours of service per year to their community. The hall has an average attendance of 600 visits per week. Entry to events is paid in time money or in conventional money: A two-hour bingo night or movie costs two time credits, and a three-hour cultural performance is valued at three time credits.⁵

People learn to cook healthy meals in the community food studio and serve it in the café, which prides itself on offering food grown locally in community and individual gardens.

Besides the economic benefits that are cultivated through local currencies, people interact in new ways, creating stronger ties between themselves and their community, transforming crime rates, antisocial behavior, and the general down-and-out demeanor of the place.

CIVICS AND NONPROFITS

Through the lens of cooperative currencies, it becomes evident that there is a new and empowered role for NGOs and nonprofits. These entities traditionally are strapped for resources and often find themselves in vigorous competition with each other for a very limited pot of money and support. Consequently, large swaths of time are spent trying to raise capital, while everybody would really prefer to focus on the organization's core mission. Salaries are usually not competitive in comparison with the commercial sector, which can make hiring and keeping the right people problematic.

Interestingly, in a society with a dual currency system in place, nonprofits not only can play a key role, but also gain access to many more financial resources.

The example of the Civic and its emergent role in government was explored in the last chapter. Such systems provide opportunities for NGOs by playing the same role in the cooperative economy that corporations play in the conventional economy: initiating projects, organizing activities, and coordinating and motivating people. Envision also a specially created new type of nonprofit that would be in charge of auditing all the other nonprofits, ensuring transparency and trust in the system. It would take on the role that auditing firms play in the competitive economy, and its reports would be published on the city's Web site. The benefits to nonprofits would be multiple: a powerful new way to reach their objectives, a big increase in their volunteer base, and the opportunity to pay their staff partially in civics.

This generic model can be customized to fit the specific objectives and needs of any given NGO or nonprofit organization.

FREE CLINICS AND ITHACA HOURS

Health care is a critical issue in the United States. Approximately one in four adults do not have health care insurance coverage. Paul Glover,

creator of the community currency Ithaca HOURS, offers innovative solutions to a wide variety of issues, including most recently health care. It all started back in 1991, with a local paper currency experiment in his hometown of Ithaca, best known for its Ivy League school Cornell University, in upstate New York. Initially, the primary focus of this cooperative currency system was to address underemployment among the townspeople by connecting them to one another in a skills directory.

Ithaca HOURS is a paper currency that comes in six denominations. Each Ithaca HOUR is worth one hour of basic labor. Thousands of individuals and over 500 businesses have earned and spent HOURS. They have made millions of dollars worth of trades with HOURS, representing hundreds of job equivalents at \$20,000 each. Businesses accepting the local currency include the Ithaca Health Alliance, Cayuga Medical Center, Alternatives Federal Credit Union, the public library, many local farmers, movie theaters, restaurants, healers, plumbers, carpenters, electricians, and landlords.

Paul Glover comments, "Ithaca HOURS is local tender rather than legal tender, backed by real people, real labor, skills, time, and tools. Acceptance of the currency is voluntary, and it is interest-free. These are two prime differences between national and local money. Approximately 11 percent of HOURS are issued as grants to community organizations. Over 100 nonprofits have received grants in HOURS. Five percent of HOURS may be issued to the system itself, primarily paying



A half hour denomination of the Ithaca HOURS. Notice the logo "In Ithaca We Trust."

for printing HOURS. Loans of HOURS are made to local businesses, with no interest charged. These loans have ranged from \$50 to \$30,000 in value.”⁶

At the height of Ithaca HOURS’ popularity, some 80 communities around the United States conducted their affairs partly with local currency. Those that thrived did so by hiring a full-time networker to facilitate circulation.

Against this backdrop and many years of experience, Glover’s latest initiative is the Patch Adams Free Clinic in Philadelphia. The American physician Hunter Doherty “Patch” Adams is involved in the project, which uses humor and play to heal people. His life story was the basis of the popular 1998 movie *Patch Adams*, starring Robin Williams. Since the 1990s, Adams has endorsed the Ithaca Health Alliance (IHA), founded by Glover as the Ithaca Health Fund (IHF). The Philadelphia clinic is the first clinic beyond Adams’s Gesundheit Institute in rural West Virginia to bear his name. The clinic will locate in a low-income neighborhood that has 250,000 uninsured residents. Unemployment and crime rates are high. Remarkably, this clinic will be member-owned and operated.

Glover explains, “The clinic will provide community-based health care that is genuinely nonprofit, preventive, humane, and, in the spirit of Patch Adams, fun. It is a refuge for doctors and nurses who want time to heal patients. It is a refuge for patients who want to be treated with dignity. For a small annual fee, members will own this clinic, gaining diagnosis and referral, dentistry, chronic and urgent care, counseling, pediatrics, birthing, hospice care, massage, family planning, chiropractic, acupuncture, and other therapies. “More than a health facility, the clinic is an economic development model that solves several urban problems. Existing medical facilities are overcrowded and underfunded. Suffering is untended, both mental and physical. Infectious disease rates are high.”⁷

Glover sees this clinic as a cornerstone of a healthy community and an economic engine to drive the area back to financial viability.

The clinic will be created using the earthship model of constructing passive solar buildings from used materials, in this case, discarded tires filled with earth. Community gardens around the buildings will grow food for the clinic.

“Expenses are kept low by relying least on U.S. dollars. To the maximum possible, we rely on the gift economy, barter, and credit systems such as Time Dollars and HOURS (‘MediCash’). Staff members are primarily volunteers: professionals, students, religious congregants, neighbors, and members. Membership may be paid with labor to maintain the facility and its grounds. Both volunteers and staff may be rewarded with Philadelphia MediCash, (HOURS) gift certificates, health care, sweat equity credits, college course credit, and scholarships. Barter agreements will also meet personal needs,” he adds.⁸

FIVE BILLION DO NOT HAVE ACCESS TO THE INTERNET

Inspired by the Curitiba example, a young American entrepreneur from Los Angeles, Kosta Grammatīs, has conceived of a new currency that links community service in developing countries with donated time on the Internet.

“Our nonprofit, A Human Right,⁹ is building a bandwidth bank where unused satellite, cellular, and fixed-wired Internet access that is donated by telecommunications companies will be put to work for social and humanitarian causes. Access to the Internet is a luxury in many countries. Currently, 5 billion people do not have access to the Internet. Although we aim to give everyone the opportunity to get online, there are many ways to go about doing this in a way that honors the access and ensures its sustainability,” says Grammatīs.¹⁰

He explains that by getting involved in community projects such as garbage collection and a variety of renovation projects, people can earn a local currency that, in turn, is good for Internet access. As a *Time* magazine feature noted: “One of the surest signs that you’re in a



Kosta Grammatis with the Echostar 16 communications satellite at Loral headquarters in Palo Alto, California. *Photo credit: A Human Right.*

developing country is the trash beneath your feet, which has less to do with bad habits than the fact that arranging garbage pickup and disposal is a low and expensive priority for a poor government.”

Grammatis adds, “Users and their contribution to their community will be tracked. Solid metrics will gauge the direct impact of community service (one ton of trash gathered for 2,000 hours of Internet) and also the social impact of Internet access (2,000 hours of Internet were used to address the long laundry list of tasks needed to be undertaken in their neighborhood).”¹¹

FUREAI KIPPU

At his retirement in 1991, Tsutomu Hotta, a highly respected former judge in Japan’s Supreme Court and minister of justice, decided to focus his energies on the growing issue of elderly care in Japan. Japan

has the fastest-aging population in the world. For this purpose, he created the Sawayaka Fukushi (Sawayaka Welfare) Center, which later became Sawayaka Fukushi Institute. Hotta's idea was to entice 12 million volunteers with the *fureai kippu* ("caring relationship tickets"), which were officially launched at that point.

The catalyzing event was a powerful earthquake that hit the Kobe area in January 1995. The capacity of the Japanese government to intervene and assist during an event of this scale was clearly overstretched. Consequently, a grassroots volunteer movement sprang up to provide help with the wide range of needs that arose during this crisis. Hotta started two significant initiatives. First, he became one of the key lobbyists for introducing the first legislation for nonprofit organizations in Japan. This lobbying was successful, and new legislation was enacted in 1998.

He also was instrumental in launching a major symposium that took place in September 1995 in Osaka for the promotion of volunteer organizations, focusing on time-based cooperative currencies. The *fureai kippu* currency and movement came out of this meeting.

Rui Izumi, an associate professor at the Senshu University School of Economics in Tokyo, points to government support as a major contributor to the growth in cooperative currencies: "The central government and many local governments are supporting local currencies in positive ways. For example, they have given financial support to some organizations, and [both] the Minister of Economy, Trade and Industry and the president of the Bank of Japan have made several encouraging remarks publicly about these systems."¹²

A former minister for economy and industrial policies described government support for complementary currencies in surprising terms: "The use of cooperative currencies can bring an end to the long-lasting deflation of the Japanese economy by supplying additional monies of various types at the local level."¹³

Government-sponsored projects were implemented in various communities of very different scales, ranging from small villages to entire prefectures (roughly equivalent to a U.S. county) and involving millions of people. All of them were designed to link unused resources

with unmet needs within neighborhoods, districts, and networks. These trials were initially run for a period of three years or so, and the results were carefully observed. The purpose for this massive investment in complementary currency experiments, estimated to be well over \$10 million per year, was and still is to fully understand what works best and under what circumstances.

The fureai kippu system provides elderly or handicapped people with any service not covered by the official national health care program. Its units are accounted for in hours of service. Different kinds of services have different valuations—for example, bathing an elderly person is given a higher hourly rate than shopping for an elderly person. In what amounts to a health care time-savings account, caregivers who provide for the elderly in the fureai kippu system accumulate credits and may draw on such credits in a variety of ways. They may use these credits themselves if they are ill, or they may elect to electronically transfer part or all of their fureai kippu credits to parents or relatives who require care and may live in another part of the country. Electronic clearinghouses perform these credit transfers. Such options ensure that ever more people are cared for.

The elderly themselves prefer the services offered by these caregivers to those paid in conventional yen currency because the relationship is different—it is more personal on both sides. Caregivers themselves often perceive the elder recipients as surrogates for their own parents.

An estimated 387 fureai kippu systems are now operational throughout Japan.¹⁴ The economic savings are substantial. As in the case of Curitiba, there is no need to raise taxes or divert funds from other programs. The human support network it creates makes it possible for the elderly to stay in their own homes longer or return home sooner after a medical intervention. The human interaction involved, along with the greater sense of community that is engendered, benefits all.

The city of Sankt Gallen, Switzerland, is the first municipality to implement a Japanese-style fureai kippu outside of Japan. During the feasibility study for this project, it turned out that only one significant change needed to be made to the Japanese model. Citizens insisted that the city

would have to guarantee that services would be provided in a different configuration if the project failed in its current iteration. The city provided that guarantee and the system is now becoming operational.

EMERGENCY RELIEF CURRENCY

Hurricane Katrina, which destroyed the U.S. Gulf Coast, China's Sichuan Province 7.8 earthquake, the devastating effects of the flooding of the Mississippi River hinterland, and the nuclear meltdown of the reactor in Japan have provided enough opportunities to learn how to better address the plight of displaced people and their communities in extreme circumstances. This usually involves a constellation of NGOs and other government agencies.



Meeting villagers in their new home, an emergency shelter, after the Yogyakarta Earthquake of 2006 (Stephen DeMeulenaere, second from left). The cooperative currency implementation project continued, although many villagers lost their homes in the earthquake. *Photo credit:* Stephen DeMeulenaere.

The following idea was originated by the Social Trade Organization (STRO), to which Stephen DeMeulenaere was one of the contributors,¹⁵ and provides the framework for a local currency that would help manage community resources more effectively during the reconsolidation phase and sustain the community once volunteers and aid organizations have left. Called the BONUS, it is a local voucher, an emergency cooperative currency designed to maximize the efficacy and productivity of relief funds by encouraging money to remain in the local economy, whose revitalization is essential for recovery. Ordinarily, with typical disaster relief programs, funds flow from donors to the local NGOs and eventually into the hands of individuals and businesses. Goods and services, however, are inevitably purchased from nonlocal sources, reversing the flow of money back out of the community. Eventually, almost all of the money leaves the area, abandoning the local NGOs to search for longer-term funding.

The BONUS strategy attempts to counteract this cycle by introducing a local voucher currency that can be used only within the community. In this system, national money is used for resources that absolutely must be purchased from outside suppliers, and everything else is purchased in the local currency. Each voucher credit is backed by a unit of national money, which allows the exchange of local currency back into national currency at any time, thus increasing the acceptance of and confidence in this currency as a valid medium of exchange. Ideally, the money would be used in the form of extremely low or zero percent interest loans payable in local currency, further increasing the acceptance of voucher money by individuals and businesses.

The effect of the BONUS program is, in essence, very similar to the effects of the saber, only in this instance instead of a learning multiplier, there is a “reconstruction multiplier,” dramatically increasing the positive impact that donors or implementing organizations have in a disaster relief situation.

Besides increasing employment, economic activity, and local access to goods and services, emergency currencies also have the potential to

reduce the amount of fraud, which is endemic in cases of crisis and common in many disaster relief situations. For example, the Government Accountability Office (GAO) estimates that 16 percent of FEMA's assistance for housing and emergency provisions in the Katrina aftermath was misspent, leading to a total loss of between \$600 million and \$1.4 billion to fraud. One man in Texas actually used FEMA money to buy a diamond ring worth over \$3,700.¹⁶ By issuing local voucher currency, the likelihood of money being spent fraudulently is significantly reduced, as goods and services can be purchased only from known vendors within the local economy.

Although the BONUS system does not explicitly endorse a demurrage for its currencies, it is suggested that any emergency relief currency carry a small demurrage fee to promote its continuing circulation, as well as to counteract the hoarding mentality that is all too common in the aftermath of large-scale disasters.

Stephen DeMeulenaere comments, "This blueprint was used very successfully by the French agency Triangle, using my design for a simplified revolving microfinance circle in Aceh, Sumatra, in Indonesia following the 2004 tsunami. And emergency currencies were used in the recent Haiti disaster."¹⁷

Perhaps Edgar Cahn, founder of TimeBanking, puts it best. Reflecting on the power of cooperative currencies to create community, no matter if they're organized by a business, NGO, or government, he commented, "Rebuilding community takes creating organic networks of trust, reciprocity, and engagement. Sustaining that is vital. A way to bring energy and vitality to that effort would be for a gathering to ask, 'What kind of memories would I like my child to have about growing up in this neighborhood? What are the stories we would like them to tell their children and friends about something memorable they did growing up in that particular place?' Suppose they could say, 'We had a TimeBank or cooperative currency that helped veterans' families and made sure that no third-grader went on to fourth grade without being at a third-grade reading level and having a buddy or mentor as

they passed to fourth grade.’ Suppose it was a tree-planting campaign as a way of reducing the carbon footprint. Why not have a brainstorming session to create place-based memories?

“Set up a TimeBank Rotating Loan Club wherever anyone who has a cause, a dream, a vision of what they would like to change or what they would like to see happen could get a time commitment of 90 days to make that happen. Like Rotating Loan Clubs, everyone would have a chance to do that: I’ll help with your cause for 90 days if you will help with mine. Imagine the possibilities.”¹⁸

PART THREE

RETHINKING MONEY

In the process of rethinking money, mistakes and accidents have occurred, some by design, some by happenstance. Whatever the causes, there have been some dire consequences. Individuals who have questioned the orthodoxy of modern-day money have, for the most part, been marginalized and ridiculed. Extraordinary successes that generated unprecedented prosperity, such as the local currencies that sprang up in Europe during the austere years between the two world wars, were dismantled. In more recent history, cooperative currencies have failed due to ignorance of the participants themselves, as they lack awareness of the hidden dynamics of money.

So what can be learned from the past so that we are not obliged to keep repeating our mistakes in the future? And what would a truly cooperative society, which has not existed since the dawning of the Modern Age, look like? In the following chapters, we will explore this available future that stands in such stark contrast to our world today.

Success, in the words of Winston Churchill, “consists of going from failure to failure without loss of enthusiasm.” The greatest barriers to success so far have been the fear of failure and the failure to learn from mistakes.

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Chapter Ten

TRUTH AND CONSEQUENCES

Lessons Learned

*In dreams begin responsibility.*¹

WILLIAM BUTLER YEATS,
20th-century Irish poet and playwright

It was hard to contain the emotions that were surprisingly welling up inside while I was standing on the bridge in the small Tyrolean village of Wörgl. The bridge was so different from how it had been described in various books and articles. It seemed in real life more diminutive, plainer, and definitely shorter, yet its impact was unexpectedly overwhelming. Back in the dreary days of the 1930s Great Depression, this nondescript yet iconic overpass symbolized the dreams of full employment and a decent standard of living for all. Scholars, government officials, and thousands of others traveled to this Austrian community to personally witness and learn from the miracle of Wörgl. Today, the town has little significance, noted mostly for its railway junction connecting the line from Innsbruck to Munich with the inner-Austrian line to Salzburg. A small museum run by volunteers bears homage to Wörgl's short-lived chapter in monetary history.

Here, the black-and-white photos of Wörgl's long-departed citizens going about their daily lives seem strikingly ordinary, given the backdrop

of this extraordinary moment in time. To appreciate the full panorama of what happened in German-speaking Europe in the years between the two world wars, besides the earlier example of the WIR, a look at Wörgl and the Wära provide some important insights.

Mostly forgotten today is that the large number of cooperative currencies arose in the aftermath of the German hyperinflation of the 1920s, when the Reichsmark, the German currency at the time, became worthless. Similarly, there was an explosion of local currencies in both Western Europe and North America following the economic crash of 1929 and, more recently, in Argentina, following the collapse of its national currency in 2001. And now, at present, there is a resurgence of cooperative currencies and other innovations as the shadow of recession looms, but the dire consequences and tough lessons from these experiences seem to have lapsed from memory.

Back in the day when Michael Untergugenberger was elected mayor of Wörgl, in 1931, some 30 percent of the workforce was unemployed, leaving 200 families absolutely penniless. The mayor-with-the-long-name, as the renowned U.S. economist Irving Fisher from Yale would call him, was familiar with Silvio Gesell's work. A German economist and merchant, Gesell's conceptual framework for demurrage and for other theories made him, some argue, the grandfather of modern-day cooperative currencies. His monetary designs are often referred to as *freigelt*, or "free money" in English.

The mayor decided to put Gesell's ideas to the test, as there was much to be done around the town and many willing and able-bodied folks looking for work. The rub was, however, that there were only 40,000 Austrian shillings remaining in the bank, just enough to pay the salaries of a couple of dozen people for one month, far short of what was needed.

Instead of spending his entire budget, which would have been gone in a matter of weeks, the mayor decided to put the money on deposit with a local savings bank and issue *labor certificates*. The deposit served as a backing or guarantee for the town's own cooperative currency, which became known as simply the Wörgl. Untergugenberger's idea was to get the town back to work and the economy moving

again. So he designed the Wörgl currency to function solely as a medium of exchange. To that end, a demurrage charge was applied through a stamp affixed each month at 1 percent of face value. Like all other demurrage charges, this *relief tax* acted as an incentive to keep the currency in circulation. Everybody who was paid with the Wörgl made sure he or she spent it quickly before the stamp's date expired.

Interestingly enough, only the railway station and the post office refused to accept the local money, most likely because they were part of the national government. When people ran out of places to spend their local currency, they would pay their taxes early, resulting in a huge increase in the town's revenues. Over the 13-month life span of the Wörgl, the town council not only carried out all the intended works projects but also went on to build new houses, a reservoir, a ski jump, and the famous bridge. The people also used their cooperative currency to replant forests, planning ahead to generate revenues from felling mature trees for timber.

Alex von Muralt, who made an in-depth investigation of the Wörgl at that time, reported the mayor's delighted comments that "taxes



People in line to cash in their Wörgls.

were eagerly paid,” sometimes even in advance. Von Muralt concluded, “This eagerness to pay taxes may be, in my opinion, simply owing to the fact that the businessman who finds, at the close of the month, that he holds a considerable amount in relief money [Wörgl], can dispose of it with the greatest ease and without loss by meeting his parish [local tax] obligations. A change of attitude has manifestly taken place. If formerly the paying of taxes was deferred to the last, now it occupies first place.”²

Wörgl soon became the only town in Austria with full employment. As pointed out in Chapter 5, local currencies are exchanged more frequently than conventional money because the latter is used for savings as well as exchanges. It is estimated that the Wörgl stamp scrip circulated in transactions from 12 to 14 times more frequently than the national currency,³ a major factor in the town’s recovery.

According to Irving Fisher, “Free money may turn out to be the best regulator of the velocity of circulation of money, which is the most confusing element in the stabilization of the price level. Applied correctly it could in fact haul us out of the crisis in a few weeks. . . . I am a humble servant of the merchant Gesell.”⁴

Word of the success of this cooperative currency spread like wild-fire. French Prime Minister Edouard Dalladier even made a special visit to see for himself the miracle of Wörgl. Soon more than 200 other towns and villages in Austria wanted to use this system as well.

It was at that point that the Austrian Central Bank panicked and decided to assert its monopoly rights by making it a criminal offense to issue Wörgls. The town of Wörgl, almost overnight, returned to 30 percent unemployment.

THE GERMAN WÄRA SYSTEM

By 1923, the monetary situation in neighboring Germany was wretched. Prior to World War I (1913), one U.S. dollar was worth 4.2 German marks.⁵ By the time inflation peaked in November 1923, the exchange rate for one dollar was 4.2 trillion marks. A postage stamp cost bil-

lions, a loaf of bread required a wheelbarrow full of money, and 92,844,720 trillion German marks were in circulation.⁶ Daily wage negotiations preceded work. Salaries were paid twice per day and were spent within the hour.

Based on Gesell's teachings, an organization was formed in Germany in 1929 to "fight stagnation of the market and unemployment."⁷ Called the Wära Tauschgesellschaft (the Wära Trading Company),⁸ it issued demurrage-charged *wära* bills. The term *wära* is a combination of the German words *ware* ("goods") and *währung* ("currency"). Wära, demurrage-charged scrips, were circulated for a period of one year, at the end of which they could be traded for new ones.

The wära, like the Wörgl, drew international attention when it was used by a local entrepreneur to help save the coal mine in the small town of Schwanenkirchen. Like many other businesses in Germany during this severe economic downturn, the mine had filed for bankruptcy and was shut down. Max Hebecker, the former production engineer, had a chance to purchase the mine for 8,000 Reichsmark, far below its estimated value. As Hebecker had no access to a bank loan, however, he decided to apply the concept of the wära. He gathered the workers and explained that the coal mine could be reopened and they could work again if they were willing to be paid in wära. This currency would be backed by the coal they were extracting, rather than national currency. A predictably lively exchange between the miners, villagers, and local shopkeepers followed. In the end, all agreed to accept payment in the new wära currency.

The wära stamp scrip not only saved the coal mine and the town but also began to circulate nationally. Soon, over 2,000 corporations throughout Germany began accepting and paying each other with this local money. A growing number of banks even opened wära accounts.

Here again, the Reichsbank, Germany's central bank, felt threatened by the wära's enormous success and by other local currencies in circulation during the 1920s. With the enlisted help of the government, these cooperative currencies were declared illegal in October 1931 through the "Brünningsche Notverordnungen."⁹

RISE OF THE NAZI PARTY

The repression of cooperative currencies, together with other anti-inflationary decisions by the Reichsbank, led to a sharp decline in the German money supply.¹⁰ As a result, the Schwanenkirchen mine and hundreds of other businesses were forced to shut down, and unemployment soared again. As helping themselves on a local level became increasingly more difficult for people, advocates of centralized solutions gained appeal. In the beer halls of Bavaria, an obscure Austrian immigrant began drawing audiences to his fiery speeches that promised a return to jobs and glory. His name was Adolf Hitler.

What was Hitler's formula for political success? No one knew better than Hjalmar Schacht, chairman of the Reichsbank and considered the leading German central banker of his generation.¹¹ According to Schacht, the Nazi Party's popularity was primarily fueled by "poverty and unemployment."¹²

The statistics invite conjecture: Between 1924 and 1930, the period when cooperative currencies were in their greatest use in Germany, unemployment hovered around 1 million people. During that same period, the percentage of seats held by the National Socialist (Nazi) Party declined from 6.6 percent to 2.6 percent. In contrast, as the cooperative currencies were outlawed and unemployment shot up by almost 500 percent in just three years,¹³ the percentage of seats obtained by the Nazi Party climbed first to 18.3 percent in 1930, then to 43.9 percent by the end of 1933, enough to bring Hitler to power.¹⁴

THE UNITED STATES

During the Great Depression of the 1930s, the success of the Wörgl experiment caught the attention of U.S. policy makers struggling with unprecedented unemployment in their country. Key conversations ensued between Irving Fisher, the Yale economics professor; Professor Russell Sprague from Harvard; and Dean Acheson, undersecretary of the Treasury. All three became convinced that the Wörgl model of-

ferred a way out of the Depression. Economist Fisher stated for the record, “The correct application of stamp scrip would solve the Depression crisis in the United States in three weeks!”¹⁵

The idea of stamp scrip was, therefore, recommended to President Roosevelt. As reported by Fisher, though Roosevelt was himself impressed, the final decision on whether to officially endorse cooperative currencies was left to advisors, who instead favored a series of new centralized programs for which political credit could be more easily claimed. These new programs included the expansion of the Reconstruction Finance Corporation and large-scale work-creation projects managed by the federal government, all part of what would become popularly known as the New Deal. Roosevelt announced by executive decree that he would prohibit *emergency currencies*, the code name given for cooperative currencies. This decree applied to any and all such currencies then in existence and any that might be proposed.¹⁶ The United States thus chose not to take advantage of such monetary innovations to address the economic crisis of the 1930s.

Contrary to popular belief, the centralized initiatives taken by the Roosevelt administration did not actually pull the United States out of the Great Depression. Certainly, the centralized initiatives and programs enacted did assist in providing valuable employment to many hardworking people. But most economic historians today agree that for the United States, as well as for Germany and Austria, it was the economic shift in preparation for, and eventual participation in, World War II that ended the Great Depression. Roosevelt himself admitted that much when he stated, “It was Dr. Win-the-War, not Dr. New Deal that ended the Depression!”¹⁷

HISTORY REPEATS ITSELF?

It’s impossible to prove that the ghastly years of Nazism wouldn’t have taken place if the grassroots currency initiatives had been allowed to thrive.

What is known is that monetary crashes invariably leave people in fear, despair, and anger. This is an explosive social mix that irresponsible demagogues can and do exploit, even today. What started as a monetary problem in the former Yugoslavia, for example—exacerbated by the IMF readjustment program in the late 1980s—was swiftly transformed into intolerance toward “others.” Minorities were used as scapegoats by ethnic leaders to redirect anger away from themselves and toward a common enemy, providing the sociopolitical context for extreme nationalist leaders to gain power in the process. Within days of the 1998 monetary crisis in Indonesia, mobs were incited to violence against Chinese and other minorities. Similarly, in Russia, discrimination against minorities was aggravated by the financial collapse of the 1990s. With the fall of the Berlin Wall and the collapse of Soviet communism, it could be argued that the identified archenemy of the United States has now been supplanted with a new foe, immigrants and the poor. Today in the United States, there are some 1,018 identified hate groups compared to *only* 602 in 2000.¹⁸

Another important lesson, and an expensive one in terms of human misery with regard to cooperative currencies, is revealed by the more recent economic crisis in Argentina.

THE ARGENTINE CREDITOS

As a model debtor nation, from 1991 to 2001 the Argentine government adopted all the policies suggested by the IMF, including privatization of government assets and a fixed parity between the Argentine peso and the U.S. dollar. Instead of leading to economic stability, however, these policies created an overvalued peso and a massive economic contraction.

People at the grassroots level responded to the economic tightening with several mutual aid initiatives. In 1995, the first *trueque* (“barter”) clubs were started to enable groups of friends and neighbors to exchange goods and services. Trueque clubs soon spread throughout the greater Buenos Aires region and into other provinces. Various clubs

then began to issue their own *credito* currency notes, and, by early 2001, there were several dozen currencies in circulation. An informal network, the *Red Global de Trueque* ("Global Trading Network"), allowed the different systems to interact.

In December 2001, Argentina went into a financial meltdown: Banks closed for months, and people could not access their accounts. The peso suffered a massive devaluation, and chaos reigned. Without the availability of cash, huge numbers of people joined the trueque clubs to make ends meet, and the amount of trading in credito currency exploded. By July 2002, an estimated 7 million people were using cooperative currencies on a regular basis. By November of that same year, however, the trueque movement had shrunk back to about 70,000 participants, roughly a 99 percent drop.

Sergio Lub is an entrepreneur born in Argentina and the founder of thank-yous, a global reputation currency used by the Friendly Favors network. Lub has traveled back to his native land frequently over the



Club de Trueque in Buenos Aires, 2001 Zona Oeste. Photo credit: Sergio Lub.

years and is very familiar with this chapter in its history: “The Papelitos system worked very well in the beginning. In 2001, when Tom Greco and I visited their clubs of trueque, they had hundreds of vendors who alternated as customers. They call themselves ‘prosumidores,’ a combination of ‘producer’ and ‘consumer.’ One participant could bring tomatoes from her garden and return home with ravioli for dinner, a voucher for a prepaid dentist visit, furniture, or fishing equipment, and all these goods and services exchanged hands at a dizzying speed using these crudely made papelitos, a substitute for money that people improvised to stay alive when banks closed and there was no more legal tender. They collectively have broken the illusion that people could not issue money on their own.”¹⁹

Heloisa Primavera, a cofounder of over 200 barter clubs recalls, “We started off as part of a group of teachers and researchers in 1996. We were working with socially at-risk groups, such as homeless children, prostitutes, and former convicts, and trying to promote self-esteem and small jobs in the face of growing unemployment due to the structural adjustment plans imposed by the International Monetary Fund (IMF) to debtor countries, especially in Latin America. One of our activities was to train people to form knowledge exchange networks in which every member was supposed to both learn something and teach something else. This strategy came from the French group Réseau d’Échanges Réciproques de Savoirs. We created then the Economic Literacy Program, and the project expanded by leaps and bounds.”²⁰

Lub and Primavera see the reasons for the failure of the trueque movement as twofold: The value of the credits became highly inflated because the money was fiat-based and run by a small group of leaders. Furthermore, the system, for the most part, was not transparent to its users; therefore, organizers were able to keep their actions secret.

“Another major problem was that the credits were primarily created as paper currency, without adequate safeguards against forgery. For example, in 2002, outside one of the big fairs, a couple of men were selling papelitos at a huge discount; you could buy \$50 worth of credits for \$1 of official money. They were denounced, and a sympa-

thetic judge ordered their arrest but had to release them the next day because counterfeit laws only protect legal tender. Furthermore, they were both employees of one of the large banks that had color copiers to make papelitos at will. To me it was clear that the bank, after reopening, counterfeited the people's money as a way to destroy their competition and regain their monopoly on issuing money," adds Sergio Lub.

Such an accumulation of flaws provided ample temptation and opportunity for corruption, which led to the collapse of these systems. To aggravate the chaos, professionally counterfeited credits had begun circulating, sabotaging the system.

According to Primavera, "What we learned from this incredible experience in terms of governance is that money is a key aspect of democracy, that it is possible to distribute wealth instead of concentrating it, if we use social money. But we also learned it is very hard to sustain such a system, if a rigorous management is not available to promote permanent transparency of accounts and decision-making processes. Still, much deeper than the technical issue, we consider the aspects of human behavior that could explain the significant growth of Argentinean networks, which occurred exponentially during almost seven years: How was it possible that a group of 23 people became 2 million in seven years? Where can you find such an example anywhere in the world, without any external support? With only self-management of groups?"

Lub reflects, "Ironically, we had already a transparent accounting system online with our thank-yous that could provide all the safeguards they needed to move their accounting online, where it would remain transparent and cannot be counterfeited (unlike cash transactions, every online transfer has to have a buyer and a seller to be recorded, so crooks cannot be anonymous). I offered our application as a gift to the leadership of one of the largest clubs, but the users were not computer literate and they were in a survival mode, unable to concentrate enough to learn how to access the Internet and make a transfer. So another important lesson for me was to do the preparations and training *before* the crisis starts. Today, I foresee the interface to be easier, thanks to cell phones, so people can access their accounts and

make payments through their phones. I saw phones used to buy sodas in 2002 in Japan, and in South Africa in 2005, so eventually we shall have it in the United States as well.”²¹

ROTTERDAM

Curiously, even well-designed and successful currencies have hit the rocks. In Rotterdam, the Netherlands’ second largest city with the largest port in Europe, over 10,000 people participated in an alternative currency in 2001. The *NU-card* experiment was designed specifically to advance a long list of green activities, from taking public transport to recycling or volunteering. Green products could be purchased using *ooins* in one of the 100 participating shops. The currency was good at the zoo and in cinemas, theaters, museums, and other recreational outlets. It was backed by the European Union and one of the largest banks of the Netherlands.

The program was stopped after 18 months because the political climate in Rotterdam changed dramatically, following the assassination of Pim Fortuyn, an ultra-right-wing politician. Fortuyn’s party won half of all the seats in the city council in the elections that followed shortly after his death, and on assuming power, it halted all environmental projects in the city.

“What we learned from this experience was invaluable. It is critical that all key players are stakeholders in the process,” Edgar Kampers submits.²²

It would be wise to heed history’s hard lessons. This is even more important today as there are greater possibilities for successful applications, given the convergence of new technologies, more access to information, and the maturity of local and regional currency system designs. Additionally, governments are learning that affording greater opportunities to their citizens bodes well for society and avoids the circumstances that give rise to civil unrest and turmoil.

So what are the design features that bode well for a sustainable currency strategy?



Chapter Eleven

GOVERNANCE AND WE, THE CITIZENS

An Ancient Future?

*The economy of the future is based on
relationships rather than possessions.*

JOHN PERRY BARLOW, former lyricist for
the Grateful Dead and founding member of
the Electronic Frontier Foundation

Bali is a small island in the Indonesian archipelago that fervently embraces and preserves its Hindu culture, though it is situated in a huge, mostly Islamic nation. What makes Bali compelling is the longevity of its complex cooperative currency system, which is inextricably interwoven with its cultivation of rice, allocation of vital water rights, celebrations of festivals, and hyperdemocratic system of governance.

From this ancient civilization, there are gripping and undeniable guidelines and strategies that may have some useful lessons relevant for our collective future.

The community at large is organized into three main networks:

- *Banjar*: the most important civic organization, which is in charge of the social aspects of the community
- *Pemaksan*: a group responsible for the coordination of religious rituals
- *Subak*: water irrigation cooperatives for rice production

John Stephen Lansing, a professor at the Santa Fe Institute whose research explores the ecology, common property, and social theories of Bali, as well as integrative modeling of environmental changes, explains that an agrarian system of rice cultivation flourished there for almost 1,000 years.¹ Then, in the 1970s, the World Bank insisted, as part of its financial aid package, that the “unsophisticated traditional methods” be replaced with modern techniques and know-how, coupled with the use of chemical fertilizers. The well-documented experiment was a disaster, causing crop failures and massive ecological issues, such as the depletion of large swaths of native coral reefs. It was then decided that the customary agricultural practices could be reintroduced. So the Balinese went back to producing two bountiful crops a year while managing pest control naturally.

Interestingly, what Lansing and his associates demonstrated, using agent modeling,² is that the Balinese traditional, socially complex, multitiered, hyperdemocratic methods of sharing precious water, along with the timing of planting and harvesting, mirrored their optimal scenario for the environment at large.

Intertwined with their time-honored agricultural and social practices is the longest-surviving cooperative currency ecosystem in the world. Before Indonesian independence in 1945, the conventional currency was the Dutch guilder, followed by the Indonesian rupiah. Working in parallel with this conventional money is a social currency networked through the Banjar system, which has been in use since before written records. Everyone in the Banjar is obliged to contribute to the well-being of the community by helping with jointly decided projects. The Banjar system extends beyond religious or cultural events and may include civic activities, such as support for building local roads, especially when the central government is unable to provide funding. In short, local resources are mobilized regularly to support a full spectrum of undertakings, whatever the community chooses to focus on. It is this system of mutual cooperation and assistance that accounts for the resiliency of Balinese culture.³

Additionally, a physical form of money—an odd-looking Chinese coin with a square hole in its middle called the *uang kepeng* (“coin money”)⁴—was also used. This coin money was minted in China and used as trading tokens, much in the same manner that trading beads were used in North America by indigenous peoples.⁵ This coinage was outlawed, however, in the 1950s.

“Chinese money, known also as *pis bolong* in Bali, has circulated as a local medium of exchange here for at least the past 1,000 years and only ceased to be used for local purchases in the early 1970s. Up until that time, a Balinese could use uang kepeng in many ways, from buying meat and vegetables in the market or snacks in front of the school, to watching movies with friends at an outdoor theater,” says Stephen DeMeulenaere, founder of the Cooperative Currency Resource Center and a long-term resident of Bali.⁶

According to DeMeulenaere, uang kepeng became the official medium of exchange of the Banjar and could be taxed, spent on public works projects, and circulated as a fully functioning currency. Thus its significance to Balinese society was formalized.⁷

It was only when new national banking and currency laws were put into effect, after Indonesia gained its independence, that the Balinese were forced to accept the Indonesian rupiah in place of the uang kepeng.

“The decline of uang kepeng as a medium of exchange corresponded with a shift in economic behavior toward earning the Indonesian rupiah. Although many significant elements of traditional life remain vibrant in Bali, mostly thanks to the Banjar-based social currency, the monetary protection blanket they once had with uang kepeng has been stripped away, leaving the Balinese people and society increasingly vulnerable to situations beyond their control and subject to the same financial and consumption pressures faced by all of us living in the modern world,” DeMeulenaere adds. The social currency, the *nayahan Banjar*, continues today as the mainstay of the Banjar system and has served to partially fill the void created by the banning of the uang kepeng.

The Balinese system of governance is of particular interest. A leader of any given Banjar is elected by a majority vote of members and, though this is rare, can be dismissed by another majority vote at any time. He or she receives no remuneration for this role. Anthropologists Clifford and Hildred Geertz describe the leader as “more an agent than a ruler.”⁸

Each family has one representative in the *krama*, the Banjar council, in which every member is considered equal and has one vote. No special status is granted to wealthier or higher-caste members. When addressing one another in the *krama*, they do not use the formal form of language for those of a superior class, which is a prerequisite of daily interaction outside a Banjar meeting. At monthly meetings, new activities are proposed, and ongoing projects are discussed. The contributions of in-kind and conventional money for each project are then decided on, customarily by a majority vote. In short, the Banjar functions as a community-based planning and implementation organization, and it budgets all its activities in two currencies: the national currency and the *nayahan* Banjar time currency.

Modern-day pressures on Balinese society, as in any developing country today, are increasing and whittle away relentlessly not only at indigenous culture but also at resources. Stephen DeMeulenaere adds poignantly, “Property values have as much as tripled in the past couple of years, especially following the success of the popular book and movie *Eat, Pray, Love*. The rice fields are being abandoned and sold to the highest bidder, and resorts and hotels are being built in their place.”⁹

To romanticize the Balinese experience would be a mistake, but to dismiss its many teachings out of hand would be one, too. The rhetorical question of the relevance of some small island’s curious culture could well be proffered. The answer is that they have managed their resources and culture for centuries through a multitiered, highly participatory democracy that has survived countless foreign intrusions. There is a clear distinction to be made, nonetheless, between wisdom and education. One can be a lowly farmer toiling on the lands of the Ngong Hill in Kenya or be pushing a broom in some sweatshop in Shanghai, have

little or no formal education, and be truly wise. The corollary to that also stands.

A modern governance system, although very different, reflecting the fractal nature of the Balinese system, is holacracy. Its structure and procedures integrate the collective wisdom of people while aligning a venture, network, or business with its “broader purpose and a more organic way of operating and the result is dramatically increased agility, transparency, innovation, and accountability.”¹⁰

DEMOCRACY, TRANSPARENCY, AND ACCOUNTABILITY

The key aspects of the Balinese cooperative currency system—democratic governance, transparency, and accountability—are clearly necessary conditions for a cooperative currency to succeed over time. These are the same lessons learned from the Argentinean debacle, for instance, that convinced the German regio movement to incorporate the following principles into their eight conditions for a currency to be granted the use of the regio label, according to Margrit Kennedy, the initiator for the regio movement.¹¹ These conditions state that these criteria need to be satisfied:

1. The system needs to be a win-win model for all participants.
2. Its aim is to support a public good.
3. It needs to be professionally managed.
4. Total transparency of the accounts and the mechanisms for all participants.
5. Democratic control by its users.
6. Sustainable finances or at least a sustainable financial strategy.
7. Guaranteed circulation of the currency.
8. Willingness to collaborate with other regio projects.

Finally, there is a preference that the currency does not bear interest. This is not a formal obligation, but so far all regios have been implementing that ninth principle as well.¹²

As we have seen repeatedly, economic hardship gives birth to a multitude of local currencies. The trend toward cooperative currencies will be stronger now than at any time previously because of the convergence of cheap computing, the Internet, and the technology to launch such systems. The vast majority of the initiatives that arise as a consequence of the current global depression will be started with a lot of good intentions. But good intentions may not always be sufficient. Inevitably, some initiatives will be technically flawed, some may be outright frauds, and some may even be deliberate attempts at sabotage, as appears to have been the case in Argentina.

Dee Hock is the founder and a former CEO of the VISA credit card association. Back in 1968, he convinced Bank of America to give up ownership and control of their BankAmericard credit card licensing program. The new entity was a nonstock membership corporation owned by its member banks. In 1976, its name changed to VISA. This new structure he calls a *chaord*, a word that comprises both the terms *chaos* and *order*.

He writes, “We must conceive of and help implement wholly new forms of ownership, financial systems and measurements, free of the attempt to monetize all values that bind [us] to next quarter’s bottom line, gross mal-distribution of wealth and power, degradation of people, and desolation of the ecosphere, or our stories will be increasingly immoral and destructive.”¹³

John Boik, a medical doctor who specializes in new cancer treatments, has applied his knowledge of natural systems to governance. He presents intriguing and important ideas on the functioning of an integrated cooperative currency and governance system.¹⁴ He addresses two key mechanisms of contemporary society that make it unsustainable: (1) The financial (banking and investment) system demands continuous economic growth, and (2) financial and political power is centralized within a small subset of the population.¹⁵

“The Earth has finite resources and a limited capacity to absorb the waste products of human economic activity; unlimited growth is unsustainable due to real ecological constraints. And the country suffers

from inherently undemocratic hyper-inequalities of power that cause widespread suffering and will likely lead to greater civil unrest.”

Boik contends that the current financial, economic, and governance systems require an overhaul. “Tweaks and reforms, while important,” he says, “will ultimately fail to address foundational failures on which they rest.” His proposal calls for developing demonstration projects at the local level and on a volunteer basis to test and refine sustainable financial, economic, and governance systems. These projects, called “principled societies,” are essentially local membership clubs open to anyone who wishes to participate.

In his blueprint, financial and political power is decentralized, in part through collaborative direct democracy, the governance system a principled society employs when adopting and refining rules. In this form of mass online democracy, members participate in the full creative problem-solving process rather than simply voting yes or no on proposals advanced by a small group. The system relies on built-in design elements that ensure efficiency, transparency, and effectiveness.

Boik adds, “Power is also decentralized through the democratic features built into the proposed financial system, a local currency system called the Token Exchange System.” Unique to Boik’s proposal is a crowd-funding–like mechanism, which pools small amounts of resources and money from a diverse group of people, usually via the Internet. This generates investment funds for business, as well as a steady stream of donations for funding social services such as schools, nonprofits, clinics, universities, and other groups. Members must use a certain portion of their incoming tokens to make interest-free loans to businesses and make donations to participating nonprofits. Individuals choose the programs they want to support.

Some of the interest-free loans would go to principled businesses, new socially responsible hybrids that contain elements of the nonprofit and for-profit models. This profit-neutral investment system accomplishes three goals: First, every individual participates and becomes a member of the investing class. Second, decisions are based on social concerns rather than profit. Third, in a profit-neutral investment

system, funding will tend to flow toward those businesses that provide the greatest community benefits. Principled businesses have a competitive advantage here in that they are designed to deliver social gains.

Boik adds, “Since investment is not motivated by profit, the system does not demand continual economic growth and can remain robust even under steady-state conditions. Profit-neutral investing is compatible with long-term ecologic viability. Thus, collaborative direct democracy is integrated with decentralized investment and social services funding. Each major aspect of a principled society—governance, finance, and economics—serves to distribute financial and political power to all system participants.”

Boik asserts that from its core foundations, principled societies and those that follow this plan are inherently democratic.

The first step that cooperative currency networks should urgently consider is to establish international standards of quality and security with an Internet-based rating agency that produces a *Cooperative Currency Consumer Report*, complete with a reputation rating similar to that on e-Bay’s auction site. The criteria for such a rating should include not only the technical aspects, such as how the currency is created, its delivery mechanisms, and its security, but also its scale and liquidity, its transparency, its governance, and its underlying social values. Bottom line: It should become possible for anybody to consult the rating of a currency before accepting it.

For any cooperative currency system that is destined to go mainstream, security measures should be commensurate with those of the conventional monetary system. There is one advantage that the cooperative currency domain has over the conventional system in this regard: Cooperative currencies should embrace transparency to its users and accountability. This is in stark contrast to the conventional system.

Additionally, the electronic technologies now available make it feasible to trace the use of any currency from its birth to its final retirement, making it a less attractive target for fraud, disruption, or sabotage.

Henk Van Arkel, founder of the Dutch nonprofit STRO, makes an interesting observation: “One of the reasons why systems fail is because there are too many meetings and the initiative gets bogged down. Additionally, a currency needs to be funded so the basic infrastructure and administration costs are covered. Otherwise, there’s burnout.”¹⁶ Michael Linton adds, “In a mutual credit system such as LETS, your money is as good as your word. If your word is no good, neither is your money. It doesn’t have to get overly complicated.”¹⁷

This clearly works well in smaller, more close-knit communities, which form the backbone of most local cooperative movements, but what happens when a program goes truly global?

A CURRENCY THAT MARRIES EFFICIENCY AND EQUITY

The airline business is fiercely competitive, yet airlines need to collaborate when some unexpected disruption occurs. For instance, if a plane is delayed for whatever reason, its arrival and departure gates may have to be changed. This may affect other airlines, competitors of very different sizes and financial power. The old solution was to expand the airport. But that approach is less favored since most of the world’s important airports can no longer expand their footprint—it is too expensive, and the land is simply not available.

So, as necessity is the mother of invention, something quite out of the box had to be devised. Geert Jonker, who now works as a software engineer in his native Holland, decided to tackle this dilemma.¹⁸ His starting point took into consideration psychological studies showing that collaboration requires minimum levels of equity, which means that the benefits from the outcomes should be roughly proportional to the input of the different players. Individuals care about the payoff for others besides their own and are even willing to incur personal loss to reach a more equitable outcome.

The methodology Jonker used to unravel this complex issue was agent modeling, whereby a computer simulates all the factors of choices

and opportunities between all the players and assesses the impact on the system as a whole. In this case, options and actions are replicated based on specific goals for each airline, as well as their interactions with each other.

“What we found was when conventional money was used in the time-slot exchanges, and all the airlines were fair, the results showed that the free market system actually worked well. As soon as even one airline exploited the situation, however, by leveraging its clout to obtain more than its fair share of benefits or delivering less than promised, all the airlines would also become self-centered, and cooperation broke down,” recounts Geert Jonker.¹⁹

Bottom line: Conventional money can make a market efficient but not equitable, unless all participants keep behaving fairly forever, which is unrealistic. The erosion of community collaboration and its devastating consequences were seen in the !Kung tribe story back in Chapter 3. As soon as conventional money was introduced into their highly cooperative and supportive community, their community disintegrated.

What Jonker proposes is a new system in which the airlines work cooperatively by allocating scarce gate-time slots among themselves via a currency specifically designed for this purpose. He calls this a *spender-signed* currency.

Jonker describes how the currency works: “This is a mutual credit system like LETS where people can issue their own money. Basically, the credits count as an IOU, which has the issuer’s name on it, so whoever gives that IOU will do something in repayment. It’s called a multi-issuer currency, so there are multiple parties who might issue money, as this IOU can be passed along. In the case of air traffic management, however, I made a variation of that system, creating a *spender-signed* currency. Every user in the scheme has to add an electronic signature to every credit that goes through his or her hands. So, at any moment, every credit will have all the names of all the people who have used that credit in a payment up until that time.”

In addition, every user has a reputation on a scale from one to three, with one being low and three representing high regard. The

likelihood of a currency being accepted is a reflection of how well one knows the end issuer or the lineage of issuers and the reputation of each of them.

If the issuer or several of its users are known exploiters or swindlers, that currency ends up with a lower exchange value in a free market. For example, if one were offered the choice between 10 units with a pedigree that is aligned to one's values or 10 units for the same transaction but with a history that conflicts with one's principles and ethics, which one would be preferred? One may decide, however, to take a lower-rated currency but at a lower value.

As Jonker points out cogently: "The fact is that spender-signed currency enables a group of agents to *punish* an exploiter in a robust way, that is, in a way that doesn't break down as soon as the first of the group starts to *defect* from common cooperative principles."

He continues, "What happens is that the credits that have my name on them tend to stay close, or around me, and will tend to circle around me from the people I know and then come back to me, when the IOU is redeemed, and it disappears. The upshot is that everyone has his own little economy or exchange sphere around him or her. All these circles, all these local economies, stay interconnected and form a global economy. I think that that very nicely models the social exchange as we do in real life. So I exchange socially with people I know and people I trust, and not with the people I don't know or don't trust. The *spender signed* currency enables lots of local economies around each user's own local economy, together interconnected into one big universal economy while keeping the efficiency of a normal monetary open market—a market with money."²⁰

MONEY THAT SMELLS?

The Roman historians Suetonius²¹ and Dio Cassius²² both reported a conversation between Emperor Vespasian (emperor from 69 to 79 AD) and his son, Titus. Titus was complaining to his father about the disgusting nature of a particular tax that his father had recently introduced.

It was a urine tax, levied on the disposal of urine in the famous Roman sewer system. In response, Vespasian held up a gold coin and told Titus, "*Pecunia non olet!*" ("Money doesn't smell!").

This phrase is still used today as a way of saying that money is unaffected by its source or its users.²³ What Jonker invented, by contrast, is a system of money that tracks all its users. The currency's originator and its users leave an indelible imprint so that subsequent users can express their opinion. It is now possible to have "smelly money," as a previous user's reputation may stink!

A possible application of this is described in the next chapter, with two Irish lads working out their deal on buying a car. The clincher as to whether a bargain is struck is the origin of the funds over time.

Since available information technologies are now capable of making Vespasian's statement untrue for the first time, imagine the role and effect of a currency such as Jonkers outlined in a monetary ecology. This currency could circulate in parallel with conventional money, using the existing economic framework, but it would change the economic market into one where collaboration and reputation for fairness become as important as profit and competition. The very fact that a participant is willing to use it would be a sign of its reputation rating, and it could even make corporate whitewashing or "greenwashing" counterproductive, because any dishonesty immediately shows. It just wouldn't pay to misbehave.

This type of currency assumes, as in classical economics, that all relevant information is always available to every participant in the market, which clearly is not the truth with conventional money. Mobile Internet access, however, could create a space where a lot more information could be pulled into a decision than was ever the case.

An economy of relationships is trying to emerge: An economy in which interconnectivity empowers the individual, along with his or her various communities, evolving into a more democratic, transparent, and viable economic life, enabled by various consciously created currencies operating at all levels of society, from local neighborhoods to the world at large.

TOWARD A MONETARY ECOSYSTEM

Revising our monetary systems through the inclusion of cooperative currencies fosters the possibility of building a better world, one to enjoy now and one worthy of bequeathing to future generations.

What could be some of the components of what we have described a monetary ecosystem?

That it provides *balance* between competition and cooperation, as measured, respectively, by the system's efficiency and resilience, is crucial to the long-term health of any given complex flow system. As already discussed in Chapter 4, in all sustainable systems, the optimum point between resilience and efficiency invariably lies much closer to resilience. And these two factors for resilience are diversity and connectivity.

What could elements of a mature monetary ecosystem look like one generation into the future? For example, a multitiered monetary system could consist of:

- a global reference currency
- three main multinational currencies
- some private international scrip
- scores of national currencies
- dozens of regional currencies
- a multitude of local cooperative currencies
- a wide variety of functional currencies

Of course any one individual or business would participate in only a few of these systems. Just like today, nobody uses all of the conventional national currencies in existence. This list just describes what is available as choice of media of exchange.

Let's now explore what each of the items in the currency menu could look like.

A Global Reference Currency The need for a global currency that is nobody's national currency has finally been recognized. It has taken the form of the Terra described earlier, or some variation of it, and arose from a systematization of corporate barter.

Three Multinational Currencies It has indeed become obvious that regional economic integration can reach maturity only when a single currency levels the playing field for all economic participants. A single currency is the only way to structurally guarantee a unified information field. After some serious start-up problems in Europe, various multinational alliances have evolved from the original euro. There are also multinational currency zones in Asia (triggered by a deal between Japan and China) and in the North and South Americas (around a new Amero, or federal dollar, after a reform of the U.S. dollar).

Some Private International Scrip The airline loyalty currencies were the first large-scale application of an international corporate scrip. Today, Facebook is trying to dabble in the same direction. Google may be tempted to do the same. The net result will be that several corporate scrips will be competing on the Internet, issued by the likes of Amex, Microsoft, or an alliance of European and Asian corporations. Some have created special subsidiaries, with strong and liquid balance sheets, to issue these currencies and imbue them with greater credibility.

National Currencies In many countries, national currencies will be used for a long time to come. They continue to play an important role within any country that has not joined a formal multinational currency integration system. Most exchanges continue to involve national currencies at least in partial payments, if only because they remain the official legal tender with which national taxes are being paid.

The main difference would be that national currencies would lose their monopoly as the medium of exchange. Payments could consist of a mix of national currencies, corporate scrips, or Internet currencies, even in a single transaction.

The only places where the national currencies have retained their monopoly are a few underdeveloped countries and backward dictatorships, where political control over the Internet has kept the cyber-economy completely shut out.

Regional Currencies Under pressure to reduce long-distance transport, a significant source of harmful carbon emissions, many businesses are reorganizing their supply chains into regional networks. The C3 mechanism was one of the pioneers in this area. Supply chains have also emerged around specific industrial sectors, such as food production and processing or automobile components, and, of course, the regions in German-speaking Europe.

Even in the domain of conventional currencies, this trend is in evidence. Today, 14 U.S. states, namely, Colorado, Georgia, Idaho, Indiana, Missouri, Montana, New Hampshire, North Carolina, South Carolina, Tennessee, Utah, Vermont, Virginia, and Washington, have taken action to create their own state currency, usually backed by a precious metal such as gold or silver.²⁴ In the case of Utah, for example, the Utah Legislature has passed a bill allowing gold and silver coins to be used as legal tender in the state—and for the value of their precious metal, not just the face value of the coins. Utah's bill allows stores to accept gold and silver coins as legal tender. It also exempts gold and silver transactions from the state's capital gains tax, though that does not shield exchanges from federal taxes.²⁵

Local Cooperative Currencies In reaction to economic globalization and in parallel to it, organization of currencies at the local level has become very popular. The information revolution brought about a

systematic reduction of production and service-related jobs. As jobs grew scarcer, communities created their own currencies to facilitate local exchanges among their members. Once critical mass was attained, cooperative currency clearinghouses on the Internet made it possible for members of these communities to participate in the cyber-economy as well.

Functional Currencies The field where the most creativity has emerged is special function currencies. There could be carbon-reducing currencies, energy-saving currencies, beautifying-neighborhood currencies, and currencies designed to stop overfishing in certain waters. There are systems like reputation currencies (such as seller ratings in the model pioneered by eBay) and learning currencies (the saber, talents, and alegres). The list of what is being and can be created is limited only by our imaginations and our abilities.

With all these currencies in circulation working in tandem with the conventional money, how would life be? What transformations would possibly take place? Let's take a glimpse at a possible future.